

AI and Blockchain-Based System for Visual Quality Grading and Traceability of Apples in Supply Chains

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Abstract — A blockchain-secured mobile application integrates artificial intelligence for automating the quality grading of apples. Herein, transfer learning was used to independently build two deep learning models, namely MobileNetV3 and EfficientNet, that classify fruits into predefined grades (A, B, and C) based on their color, texture, and surface defects. The performance of each was determined to see which architecture has higher accuracy and better generalization in the real world for grading apples. Thus, this mobile app allows farmers to capture or upload apple images to get instant grading. Subsequently, the grade prediction and its metadata, including farmer details, timestamp, and location of grading, are recorded on a decentralized blockchain ledger for assured transparency, traceability, and immutability. Each batch of graded fruits will be assigned a unique QR code that can be scanned by consumers to verify the origin of their fruits and give feedback. This integrated system fosters trust among supply chain stakeholders, leading to reduced fraud, better quality assurance, and increased efficiency in the fruit distribution networks. In addition, the system facilitates farmers in viewing the nearby warehouses and collection centers around them, hence allowing quicker post-harvest decisions.

Keywords —Fruit Quality Grading, MobileNetV3, EfficientNet, Transfer Learning, Deep Learning, Image-Based Classification, Color and Texture Analysis, Surface Defect Detection, Real-Time Grading, Blockchain Ledger, Decentralized Traceability, QR Code Generation, Farmer Metadata, Location Tracking, Supply Chain Transparency, Warehouse Locator, Mobile Application.

I. INTRODUCTION

Grading is a major activity of quality assessment for apples that fixes market value, consumer acceptance, and storage suitability for the commodity. Grading in orchard sites and physical markets is generally done based on apparent features such as color, shine, texture, firmness, and bruises. This practice is generally inconsistent due to human subjectivity and changing lighting conditions, besides non-uniformity in the choice of grading criteria, and leads to misclassifications and disputes over price, thereby causing loss of trust. The present paper is targeted at overcoming limitations as

mentioned above by proposing an integrated apple grading system integrating image-based classification with blockchain-backed traceability on a mobile application. The authors have fine-tuned two deep learning models, namely, MobileNetV3 and EfficientNet, on transfer learning to classify apples into grades A, B, and C using visual features. After classification and grading, metadata such as farmers' identity, timestamp, and GPS location of the commodity would automatically be recorded on a decentralized ledger and is tamper-proof. A unique QR code would instantly verify its provenance and quality, and a warehouse locator would point to cold storage warehouses located in the vicinity that can help in timely post-harvest handling.

A. Research Gap

Traditional methods of grading tend to rely on a single model, labor-controlled laboratory datasets, or centralized databases without allowing much scope for real-world applicability and traceability. Likewise, very few research efforts have focused on apple grading, particularly using lightweight to high-accuracy architectures such as MobileNetV3 and EfficientNet. Most existing systems also hardly extend real-time guidance on cold storage or collection centers in nearby areas to farmers. These lacunae thus call for an integrated approach that ensures accuracy in grading, safety in traceability, and actionable decision support for post-harvest decisions.

B. Objectives

1. Offering a reliable grading ecosystem of apples with increased accuracy and transparency, and generally boosting efficiency in the supply chain.
2. With real-time classification of apples through AI-based image analysis, our clients get accurate and consistent grading results.
3. The objectives are to securely store the metadata of grades in a blockchain ledger for immutability, traceability, and tamper-proof recordkeeping.

4. The generation of QR-based provenance identifiers, through which stakeholders can easily verify grade, source, and handling details.
5. Provide location-based warehouse recommendations to ensure better logistics, lower transit losses, and ideal distribution.
6. This reduces mislabeling and hence improves quality assurance; consumers and distributors get genuine and correctly graded produce.
7. AI-driven grading integrated with blockchain-backed traceability to build trust throughout the distribution network of apples.

II. LITERATURE REVIEW

These are certain approaches that, in the modern-day scenario, have also gained considerable attention with the augmented need for rapid, reliable, and non-destructive evaluation techniques along all different points of agricultural supply chains. Some early attempts mainly employed classical image processing and machine learning, while more recent works increasingly adopt techniques such as deep learning, transfer learning, explainable AI, and hybrid CNN–ML models for higher accuracy and real-time performance.

A. Classical Machine Vision and Machine Learning-Based Grading

Early research focused on classical image processing pipelines with SVM and ELM classifiers. Peng et al. [1] proposed a machine-vision-based grading of apples based on grayscale conversion, segmentation, and multi-feature fusion using the SVM classifier that achieved a good recognition rate. Subsequently, other apple quality prediction studies adopted a support-vector-machine-driven grading framework based on near-infrared spectroscopy and extreme learning machines. Among these, Ordinal ELM and Evidential Ordinal ELM models provided improved robustness and better uncertainty handling [13], [15].

Attribute-based grading of mango maturity has also been done using machine learning, where decision trees and TSS/TA biochemical indicators were combined to predict the ripeness with an accuracy of 91.9% [4]. Traditional ML techniques like SVM, KNN, and ANN are still applied for specific fruits like walnuts [22] and mangoes [26], using handcrafted features derived from texture, size, and external appearance.

B. Deep Learning and CNN-Based Fruit Grading

The invention of convolutional neural networks brought about a revolution in the inspection of fruits for their quality. Patel and Patil [2] further presented an improved CNN architecture incorporating spatial pyramid pooling with FIR filtering,

which resulted in fruit disease detection and SSDAE-SVM-based grading with an accuracy of 97.25%. Narvekar and Rao [3] performed transfer learning on multiple CNN backbones to grade fruits and vegetables based on their size, shape, and color features. Further works on pomegranate [5], muskmelon [7], and citrus fruits [24] have proved that CNNs show superiority in capturing visual defects and minor variations in quality.

For large multi-fruit datasets, the best performance was achieved by DenseNet-based architectures, reaching an accuracy as high as 99.67% during the evaluation of fruits for their quality [9]. Fully convolutional UNet-based segmentation methods have also been adopted, especially when defects' localization is crucial, like in oranges and other citrus fruits [14], [12]. Object detectors like YOLOv3, YOLOv5, and YOLOv8 have also been used to identify defects in fruits and fresh-rotten classification with real-time detection efficiency enhanced in orchard and industrial settings respectively [12], [17], [19], [29].

C. Lightweight and Real-Time Models: MobileNet and EfficientNet

Lightweight CNN models are frequently preferred for deployment in mobile and resource-constrained agricultural devices. MobileNetV2 and MobileNetV3 have consistently demonstrated competitive accuracy with reduced computational cost. Kona et al. [10] showed that MobileNetV2 achieved 98% accuracy for mango defect detection. For muskmelon grading, MobileNet outperformed heavier models, balancing accuracy (87%) and real-time feasibility [7].

Hybrid CNN-ML architectures with MobileNetV3 and EfficientNet coupled with traditional classifiers like SVM have shown very promising results. For dragon fruit maturity and quality grading, the results of MobileNetV3-SVM and EfficientNet-SVM were 99% accurate [6]. Apple grade classification based on feature extraction using EfficientNet together with ML classifiers enhanced accuracy further, which finally ended in an ensemble accuracy of 98.21% [30].

D. Multi-Task, Explainable AI (XAI), and Advanced Deep Learning Techniques

Recent research focuses on model interpretability and multi-task learning. FruitQ-GradeX proposed a multi-headed CNN that performs simultaneous fruit classification and quality grading with an accuracy of up to 99%, further improved by visual explanations using Grad-CAM [8]. DenseNet, VGG, and custom UNet segmentation models have also been enhanced with explainability features for robust real-time inspection [9], [12].

Comparative studies, which have assessed several CNN architectures for classifying rotten fruits, have emphasized the dependability of transfer learning and its adaptability to a wide range of fruits and defect categories.

E. Hardware-Integrated and End-to-End Automated Grading Systems

Several works explored complete integration of hardware and software for commercial deployment. Chen and Chang [16], for instance, designed a complete apple grading pipeline on a conveyor belt, with ESP32 microcontroller and deep learning for three-grade classification. Wu et al. [21], on the other hand, implemented a small farm-specific embedded grading system powered by STM32. Complete automated sorting units have also been developed for apples, lemons, and oranges using SSD-based defect detection in controlled lighting environments[19].

F. Blockchain, Traceability, and Smart Agriculture Integration

Besides grading, contemporary systems use blockchain and IoT to create a supply chain with full transparency and traceability. Liu et al. [17] presented an architecture for fruit traceability using Ethereum blockchain, YOLOv5 detection, and improved UNet-based segmentation for saliency extraction. Similar approaches underline that the use of computer vision combined with distributed ledgers prevents tampering, ensures authenticity, and maintains quality data across all supply chain nodes..

G. Review Studies and Trends in Fruit Grading Technologies

Comprehensive reviews have integrated conventional and contemporary smart vision-based grading approaches. Singh et al. [18] reviewed state-of-the-art ML/DL approaches for non-disruptive post-harvest grading of high-value horticultural crops and identified open challenges and future opportunities. Dakwala et al. [28] evaluated various CNN models for multi-class rotten fruit classification, confirming the dominance of transfer learning in agricultural automation.

III. METHODOLOGY/EXPERIMENTAL

A. Dataset Collection and Preprocessing

With this in mind, a dataset was created to capture real-world orchard and market conditions of apples for the development of an apple fruit quality grading system. Due to the peculiar patterns of texture, color gradient features, and bruise sensitivities in apples, the dataset is exclusively prepared from various apple farms, orchards, cold-storage units, fruit mandis, and open-source image repositories. It includes varieties such as Red Delicious, Royal Gala, and Kashmir apples, with variation across different levels of ripeness and natural defects. Images were taken under different light conditions, including bright sunlight in orchards, canopies of trees, indoor lighting in warehouses, and open-air market settings. The presence of changes caused by leaves and branches, soil contamination, patches of moisture, and partial

occlusion by adjacent apples is intentionally included to simulate the real world.

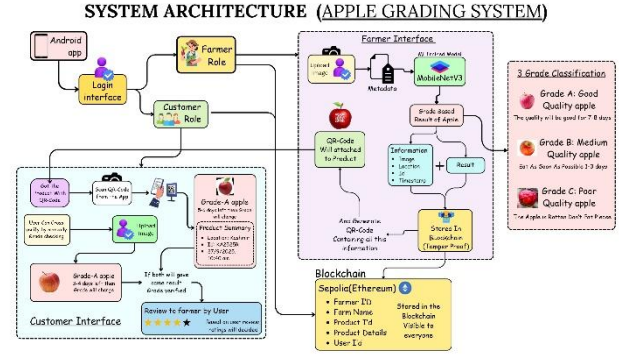


Fig. 1: System Architecture of the Proposed Apple Quality Grading and Traceability System.

Each apple image undergoes a series of preprocessing steps to maintain consistency among the images. Images are resized to fixed input dimensions compatible with both MobileNetV3 and EfficientNet; normalizations are done for balancing color intensities. As color uniformity and shine are the key indicators of quality in apples, preprocessing conserves minute details of these features carefully. Augmentations will include brightness and contrast shifts, rotation up to 25°, scaling, light Gaussian noise, zoom variation, and random cropping. Such augmentations simulate a variety of practical field scenarios: camera shake, low-light conditions within the orchard, variable positioning of apples, and partial visibility of bruising. Such variation allows models to generalize effectively when applied directly to the smartphones of farmers.

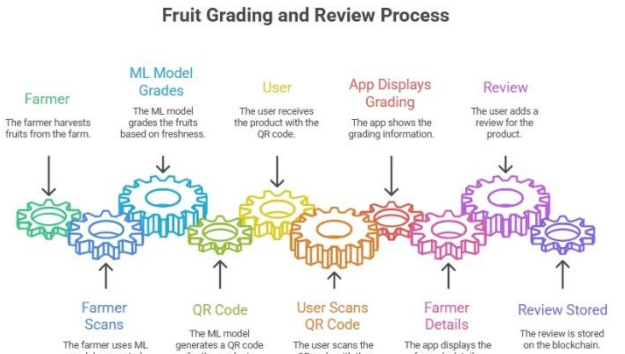


Fig. 2: Workflow of the fruit grading, QR traceability, and blockchain-based review system.

B. Model Training, Transfer Learning, and Optimization

This would be training with transfer learning on MobileNetV3 and EfficientNet, two high-precision deep learning visual classification architectures very suitable for mobile environments. Both models are initialized with pre-trained weights on ImageNet, which equips them with rich

foundational features that are necessary to detect edges, textures, shine patterns, and color gradients-all properties that are integral in grading apples.

Pre-trained classification layers are removed and replaced with custom dense layers optimized for categories of apple quality: Grade A, premium apples; Grade B, apples of moderate quality; Grade C, low-quality apples with severe defects. Then, selected deeper convolutional layers are unfrozen and fine-tuned in order to adapt generic ImageNet knowledge toward apple-specific cues such as surface glossiness, bruising intensity, skin smoothness, fungal spots, peel inconsistencies, and stem-end defects.

Optimization techniques in training that are used to make the convergence stable and avoid overfitting include learning rate scheduling, dropout regularization, batch normalization, and early stopping. The models in this research were trained until their optimum performance on the validation datasets was achieved, with their output being evaluated in terms of accuracy, F1-score, per-class confusion matrices, and misclassification trend analysis. The model with the best balance between accuracy and inference speed becomes the primary deployed model within the mobile application.



Figure 3: Model output demonstrating classification of the apple as Grade A with a confidence level of 0.80.

C. Real-Time Image Capture and Grading Workflow

After selecting the best-performing model for the grading of apples, it is integrated with a mobile application for real-time apple quality classification. Farmers will interface with this system through a user-friendly camera interface by taking pictures of apples directly in orchards or storage centers. Upon capture or uploading an image, the app will preprocess it by resizing, normalizing, and converting it into tensor format usable by the models for inference.

It then feeds this preprocessed image into the embedded deep learning model, which identifies its redness distribution, shine uniformity, shape integrity, bruises, and skin defects as the fundamental features of an apple. At the output, it gives the probability distribution across the three grades of apples, instantly pointing out the one with the highest probability to the farmer. The whole process of inference has been optimized to run under a few seconds to ensure instantaneous feedback is achieved without even relying on high connectivity at the most remote orchard location. Near-real-time grading removes the dependency on manual sorting and reduces human subjectivity considerably in the quality assessment of apples.

D. Automatic Metadata Extraction and Structuring

Concomitant with grading, the system automatically extracts critical metadata that allows downstream traceability. It includes authenticated farmer identity, the exact timestamp of the grading event, and GPS coordinates retrieved from the mobile device. The information carried in this metadata thus describes the origin of this batch of apples in terms of the orchard region where the apples were grown, the time of harvest, and finally the farm conditions.

Metadata collected then gets collated onto a structured digital object comprising grade of the apple, model confidence, orchard location, and identity of the farmer. Standardization of metadata structure makes it consistent and reliable across every batch of thousands of apples. Automation removes human entry errors and makes sure that each and every instance of grading apples gets documented correctly. The structured object is further processed to securely store it on the blockchain network.

E. Blockchain Integration, Transaction Recording, and Immutability

The system implements a blockchain ledger in a decentralized manner, ensuring full transparency and authenticity to make tampering with the documentation impossible. Every transaction of apple grading, together with the predicted grade, origin location, timestamp, and farmer identity, is cryptographically hashed into immutable records on the blockchain. Blockchain, being decentralized by design, does not allow for manipulation; hence, none of the stakeholders can alter or fake the apple quality data after recording.

Once a transaction is on the blockchain, it is transparent and verifiable to each distributed node. This engenders long-term trust along the supply chain for apples in that disputes on quality are avoided, fraudulent grade inflation is prevented, and origin tracking can be performed in a reliable way. For this reason, the ledger itself provides a secure backbone for end-to-end transparency in distributing apples.

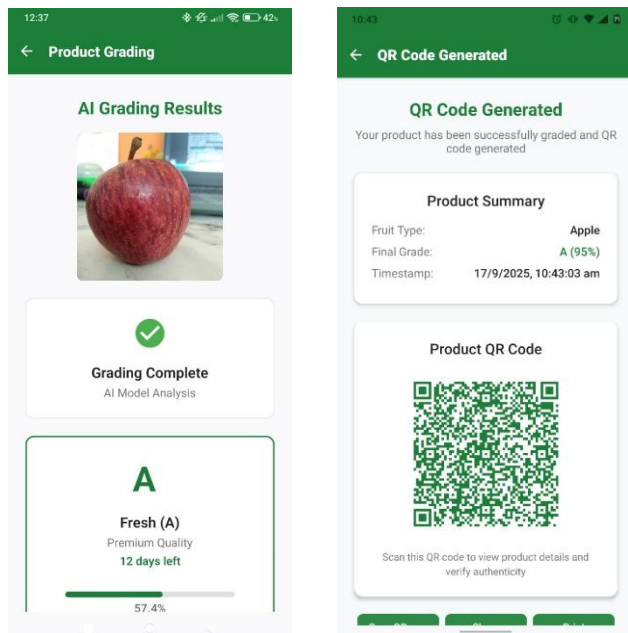


Figure 4 : AI-Based QR Code Generation Interface (right).

Figure 5: AI Grading Results Display with Fruit Quality Analysis (left).

F. QR Code Generation, Provenance Verification, and Consumer Transparency

Once the entry on the blockchain is complete, a unique QR code linked to the transaction hash is generated. This will be the digital identity for this batch of apples. A QR code is attached to the crate, storage bin, or packaging that allows retailers, wholesalers, and consumers to access full provenance information with just one scan.

A QR scan reveals blockchain-authenticated records of things like apple grade, orchard origin, grading timestamp, farmer identity, and quality proof. This instills trust right from the supply chain level, since the consumer is assured that the apple they bought is indeed of premium grade or correctly labeled according to grade. Also, this kind of QR mechanism reduces fraudulent relabeling of low-quality apples as high-grade produce and makes the market more transparent.

G. Warehouse Locator and Post-Harvest Decision Assistance

Besides grading and traceability, the system has a module on the warehouse locator that would enable farmers to efficiently take timely post-harvest decisions. Using GPS data captured during grading, the application locates apple cold-storage facilities in the vicinity, collection centers, local mandis, and procurement hubs.

It calculates distances by geospatial algorithms and then lists the nearest facilities, with estimates of travel time, including options to navigate. Apples need to be brought down to lower temperatures ASAP to retain freshness and firmness; hence, recommending the closest storage unit is a big help in reducing spoilage and losses immediately after harvest.

Overall, this integration of quality grading with logistics guidance will mean there is a smooth flow of apple batches from orchard to storage with much better operational efficiency.

H. End-to-End System Integration and Workflow Optimization

The last section shall deal with integrating all these modules, right from deep learning inference to metadata handling, blockchain storage, QR code generation, and warehouse recommendation into one seamless mobile ecosystem. In designing the system, it is modular and service-oriented, with each module exposing an interface that allows it to interact seamlessly with the next module. This way, by the time an image of the apple is captured, grading, creation of metadata, recording on the blockchain, and issuance of the QR code are done without the user's further input.

It reduces the computational overhead in mobile devices because it performs only lightweight tasks on the mobile device itself, while heavier tasks concerning blockchain operations are offloaded onto the cloud or edge servers. With this hybrid approach to processing, fast-processing performance is guaranteed; low latency and reliable execution are assured-even in rural environments. As such, the resulting end-to-end system transforms apple grading into a transparent, digital, traceable, and highly efficient workflow that supports all stakeholders across the apple supply chain, from farmers and buyers to consumers.

IV. COMPARATIVE ANALYSIS TABLE

System	Primary Objective	Key Strengths (Existing Work)	Major Limitations (Compared to Proposed System)	Unique Contribution of Proposed System
Peng et al. (SVM-based grading)	Classify fruit grades via handcrafted features	Low computational cost	No automatic feature learning; no digital traceability; metadata editable	Automated DL-based grading with tamper-proof metadata stored on blockchain.
Patel & Patil (CNN-SPP)	Detect fruit diseases	High lesion-detection accuracy	No grade classification; no secure storage; no consumer verification	End-to-end grading (A/B/C) with immutable blockchain records accessible to all stakeholders.
Narvekar & Rao (Transfer Learning)	Multi-fruit grading	Strong feature extraction & accuracy	GPU-dependent; no secure metadata; no manipulation	Lightweight CPU-friendly model + blockchain ensures results cannot be

System	Primary Objective	Key Strengths (Existing Work)	Major Limitations (Compared to Proposed System)	Unique Contribution of Proposed System
			on-protection	altered post-grading.
Kona et al. (MobileNetV2 Mango Defect Detection)	Detect surface defects	Mobile-friendly model	Only defect detection; lacks identity mapping; weak verification flow	Each fruit assigned unique QR ID + blockchain record, enabling multi-level verification (farmer→retailer).
Hybrid MobileNet/EfficientNet-SVM	Predict maturity	High maturity-prediction accuracy	No complete grading pipeline; metadata modifiable; no supply-chain linkage	Full profiling (defects + maturity + grade) stored immutably with distributed ledger consensus.
DenseNet Multi-fruit Model	High-accuracy grading	Strong representation learning	High computational load; unsuitable for edge use; no trusted audit mechanism	Real-time CPU/edge deployment + blockchain-backed audit trail for transparent grading history.
YOLOv5/YOLOv8 (Defect/Rot Detection)	Binary classification (fresh/rotten)	Industrial speed and robustness	Only binary labels; no secure audit; results editable	Multi-grade classification with encrypted hash-based blockchain storage ensuring result integrity.
ESP32 Hardware Grader	Low-cost physical sorting	Fast hardware response; affordable	Low image resolution; only local storage; vulnerable to tampering	AI grading combined with cloud-synced blockchain metadata for secure, high-resolution verification.
Blockchain + YOLOv5 Methods	Supply-chain traceability	Secure, transparent transaction logs	Not linked to actual AI grade prediction; lacks multi-level verification workflow	Unified AI grading + blockchain + QR traceability pipeline with complete verification path.

Table 1: Comparative Study of Existing Fruit Grading Methods and the Proposed Integrated AI-Blockchain Solution

V. RESULTS AND DISCUSSIONS

Experiments have been conducted to evaluate the overall performance of the proposed automated apple grading system on model accuracy, blockchain traceability, QR-based verification, and warehouse locators. The results obtained prove that the integrated system enhances the reliability and transparency of the grading and post-harvest decisions in the entire apple supply chain.

Three image classification models, namely MobileNetV3, EfficientNet-B0, and ResNet50, have been trained on the apple-specific dataset. MobileNetV3 yielded the highest accuracy of 93%, in contrast to 90.8% from EfficientNet-B0 and 89.2% from ResNet50. It required very few training images at about 2,000 images per grade, while EfficientNet-B0 required more than 3,200 images and ResNet50 more than 4,000 images. MobileNetV3 also had the fastest inference time and hence was very suitable for real-time usage on smartphones. EfficientNet-B0 was good in terms of minor defect detection but ran slowly. ResNet50 had a heavy architecture; thus, it is not suitable for mobile deployment.

Factor	MobileNetV3	EfficientNet-B0	ResNet50
Accuracy	≈93% (Highest)	≈90.8%	≈89.2%
Images Required per Category	~2000 (Lowest)	~3000–3500	~4000+
Model Size	5–6 MB (Smallest)	16–20 MB	95–100 MB
Training Time	Fastest	Moderate	Slowest
Inference Speed (Mobile)	Very Fast	Fast	Slow
Computational Cost	Very Low	Low–Moderate	High
Suitability for Mobile Deployment	Excellent	Good	Poor
Generalization (Real Orchard Lighting)	Very Strong	Strong	Moderate
Ability to Detect Fine Defects	Strong	Strongest (higher depth)	Moderate
Best Use Case	Real-time grading on smartphones	Cloud-based high-accuracy grading	Research / high-end GPU processing

Table 2: Performance Comparison of Deep Learning Models Used in the Proposed Apple Grading System

In tests conducted on actual orchard floors, farmers clicked images of apples and got the results of grades within seconds, thereby establishing the usability of the system under variable lighting and field conditions. The blockchain module created each grading record with the identity of the farmer, timestamp, grade, and GPS location as an immutable entry to ensure traceability with no tampering. QR codes

ensured that buyers and consumers accessed the verified provenance instantly. The warehouse locator also proved highly practical in identifying nearby cold-storage and collection centers, hence assuring less spoilage and better postharvest handling.

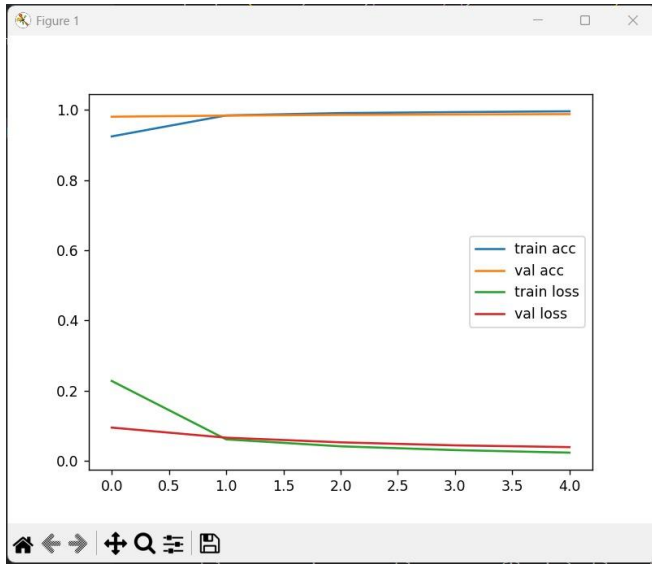


Fig. 6: Training and validation accuracy and loss curves of the fruit grading model.

The overall results thus confirm that the most efficient and accurate model to deploy was indeed MobileNetV3. The proposed system greatly enhances traceability, quality assurance, and efficiency in the distribution of apples.

VI. FUTURE SCOPE

The new apple grading and traceability system has potential for scaling up to tackle even greater challenges along the agricultural value chains. It could also be taken a step further by allowing IoT cold chain monitoring using temperature and humidity sensors, which monitor in real time the storage conditions of apples during transportation and warehousing as one measure to reduce postharvest losses. This can then be complemented with the full export-grade traceability of blockchain and QR code frameworks that allow international buyers to confirm authenticity, origin, and handling practices for better market access to farmers.

The platform can further be scaled to support FPOs by integrating smart logistics planning tools, which can optimize routes for collection, thereby cutting down on delays and enhancing aggregation efficiency. Advanced analytics integrated into the platform can provide farmers with real-time price information of the market, demand forecasts, and predictive alerts to enable better harvesting and selling decisions. All in all, the grading module itself can be expanded to accommodate other fruits like oranges, bananas, mangoes, and grapes to support multi-crop quality assessment right at the farm gate. With further enhancements related to image analysis, storage monitoring, and supply-chain intelligence, this system has the potential to evolve into a

comprehensive digital agriculture platform aimed at strengthening transparency, reducing losses, and maximizing farmer profitability.

VII. CONCLUSION

This paper proposes an automated apple grading system, an advanced image-based classification integrated with blockchain traceability and mobile decision support, to make the supply chain for apples more trustable. Of these, MobileNetV3 provided the highest accuracy with speed suitable for real-time use, trained to classify apples into standardized quality categories with higher precision. Results of grading along with all metadata like farmers' information, timestamp, and GPS location are recorded in a blockchain ledger with complete transparency and tamper evidence. Each lot of graded apples is associated with a unique QR code that would allow buyers/consumers to verify their authenticity and origin on the spot. It also helps the farmer by providing a warehouse locator for nearby cold storage and collection centers, hence reducing the chances of spoilage and enhancing post-harvest handling of the produce. In short, the solution has ensured consistency in apple grading, strengthened trust, and boosted efficiency across the value chain in the distribution of apples.

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